Windowed Multivariate LSTM Autoencoder for Leak Detection in Water Distribution Networks

Samer Ahmed[1,2*], Dilek Düştegör[1], Alexander Lazovik[1], Heinrich Wörtche[2,3]
[1] Bernoulli Institute, University of Groningen, Groningen, The Netherlands
[2] Sensors & Smart Systems, Hanzehogeschool, Groningen, The Netherlands
[3] Eindhoven University of Technology, Eindhoven, The Netherlands
[*]*samer.ahmed@rug.nl*

ABSTRACT
Leakages in water distribution networks (WDNs) are challenging to accurately identify and label, often leading to significant water loss and high repair costs. Building on existing unsupervised leak detection methods, this work introduces a Windowed LSTM-based multivariate autoencoder designed to improve the detection of incipient leaks in WDNs. We compare its performance with a similarly trained 1D CNN autoencoder using the BattleDIM dataset (DMA C), which features irregular demand patterns. The models are evaluated in terms of normal reconstruction error, factors of mean reconstruction error (a novel metric), and time to detection. Results show that the Windowed LSTM autoencoder achieves lower normal reconstruction error, higher deviation during leak events, and faster detection of incipient leaks compared to the 1D CNN baseline.

Keywords: Long Short-Term Memory (LSTM), Water Distribution Network, Leakage Detection

INTRODUCTION
Leaks in water distribution networks (WDNs) are a common and costly issue, especially in aging infrastructures. Incipient leaks often start slow and gradually grow worse over time, making them hard to notice. While catching them early can save water and money, most existing detection methods are designed for abrupt leaks (which are sudden and obvious) [1], [2]. When applied to incipient leaks, these techniques either take too long to respond [3], [4], or trigger too many false alarms, both of which can be expensive and disruptive for utilities. This highlights the need for more robust approaches specifically tailored to reduce the runtime of incipient leakages, enabling quicker detection and mitigation.

Many existing approaches rely on statistical models, but these methods often assume that the data follows a clean, Gaussian distribution. In reality, water networks data is typically non-Gaussian, seasonal, and non-stationary. As a result, statistical models often produce high false-positive rates when applied in practice [3], making them unreliable for real-time detection of incipient leaks. Supervised machine learning methods, such as Decision Trees and Random Forests, offer more flexibility and can handle non-linearities in the data. However, these models require large, labelled datasets that include both normal and leak conditions. Generating such datasets is extremely challenging in real-world water networks, as it would require highly accurate hydraulic models. Additionally, leak events are relatively rare, which leads to severe class

imbalance, further reducing the effectiveness of supervised learning approaches in operational settings.

These limitations make semi-supervised methods, and autoencoders (AEs) in particular, a promising alternative. AEs learn the typical patterns of normal operation and can detect anomalies (i.e. leaks) by measuring how poorly new data is reconstructed [5], [6]. Since they only require normal data for training, they are well-suited for environments where labelled leak data is unavailable. AEs and other neural networks can use various architectures. Long Short-Term Memory (LSTM) networks are particularly well-suited for this task, as they can capture complex temporal dependencies inherent in water time-series data, especially crucial for identifying slow-evolving leakage patterns.

However, LSTM models come with increased computational complexity, raising concerns around training time and deployability. To evaluate this trade-off, we compare our proposed LSTM-based multivariate autoencoder to a 1D Convolutional Neural Network (CNN)-based autoencoder trained under the same conditions. The 1D CNN offers faster training, lower memory requirements, and is easier to deploy in real-time systems where resources may be limited [7], [8], By benchmarking these two architectures, we aim to assess whether the improved temporal modelling of LSTMs justifies their added complexity in detecting incipient leakages, guiding utilities toward the most suitable solution for their infrastructure and operational context.

LEAK DETECTION APPROACH

We propose a Windowed LSTM-based Multivariate AE for detecting incipient leaks in WDNs and compare it to a 1D CNN-based AE, both trained using only normal operational data. The autoencoders are designed to reconstruct input data sequences. When they encounter patterns that deviate from what they have learned, like those caused by a leak, they produce a higher reconstruction error. By comparing this error to normal levels, we can identify anomalies without needing explicit labels for leaks.

We use multivariate inputs by combining data from several sensors. This adds robustness, since it is unlikely all sensors would drift or malfunction at the same time. To handle time-dependent patterns and make training more manageable, we segment the data into daily windows. This captures daily periodicity in water usage and helps the models learn more efficiently.

For evaluation, both models are tested on DMA C of the BattleDIM dataset [9], which is a hydraulically isolated part of the network with irregular demand patterns. The available sensors in this district include three pressure sensors (n1, n4, and n31), a tank level sensor (T1), and a flow meter (PUMP_1). Time series data from 2018 and 2019 is pre-processed and standardized before being split into training sets (containing only normal data) and test sets (including both normal and leakage conditions).

Model performance is assessed based on reconstruction error and time to detection. Most standard anomaly metrics are designed for singular anomalies [10], making them unsuitable for evaluating cumulative leaks. We therefore introduce a new metric: *the factor of mean reconstruction error (MRE)*, to address this limitation. This factor is calculated by taking the average

reconstruction error for a leak state and dividing it by the average reconstruction error of the normal state. A higher MRE indicates the leak is easier to detect, quantifying how clearly the model can distinguish leak events from normal behaviour.

EXPERIMENTAL SETUP

Data Preprocessing

A moving z-score standardization method [11] is applied to standardize each sensor's data. A window size corresponding to one month is chosen to account for gradual changes in flow and pressure measurements caused by seasonal variations. Failing to account for these gradual changes can result in a higher number of false positives, as flow and pressure patterns in water distribution networks (WDNs) differ significantly between seasons.

DMA C contains a persistent leak (p257) that starts in early 2018 and lasts until the end of 2019. In 2018, another leak (p31) also occurs mid-year, and is fixed. In order to include 2018 data, and therefore account for seasonality of the data, time series are selected as follows:

- first train set - from peak of p257 to start of p31 leak ('2018-01-26 00:00:00' to '2018-06-28 23:55:00')
- second train set - from after p31 leak was fixed until mid-January 2019 ("2018-08-13 00:00:00' to '2019-0114 23:55:00')
- test set - from mid-January 2019 until end of 2019 ('201901-15 00:00:00' onwards)

This method treats the p257 leak as part of the normal baseline, given that it is present and consistent throughout the selected dataset.

The multivariate data is then segmented for sequence-to-sequence model implementation. An overlapping sliding window approach [12] is used for this segmentation. A window size of 288 is selected, representing a 24-hour period with a data frequency of 5 minutes. This overlapping window structure aims to decompose the entire time series into 288 unique patterns that repeat consistently throughout the time series.

The segmentation is done for each aforementioned set, and only after this are the two train sets combined. This is done to ensure no windows exist across the edges of the two sets, as such windows would not be representative of reality. 15% of the data is randomly split from the train set to create a validation set, which is used to finetune the hyperparameters of the models.

Model Training

The two models were defined as shown in Figure 1, with the CNN having Average Pooling (size 2) and Dropout (0.2) applied after each encoder & decoder layers. Tanh activation functions were used in both Layer 1's.

The parameters in each of the autoencoders are optimized such that the Measure Squared Error (MSE) between the input multivariate data and the reconstructed data is minimal. This is achieved using an Adam optimiser with a learning rate of 0.001. The models were determined to have trained sufficiently after 10 epochs.

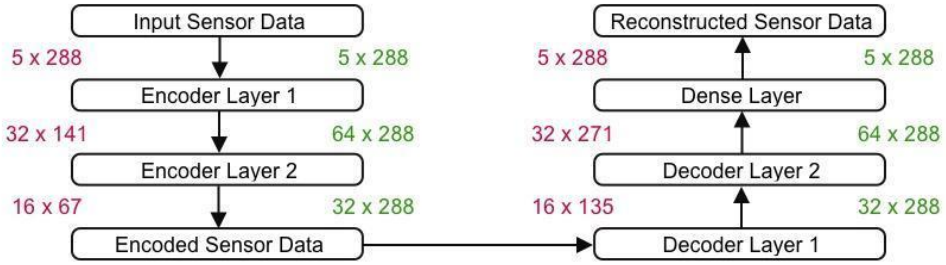


Figure 1. Dimensions of Data at each layer of the CNN (magenta) and LSTM (green) autoencoders

The test data is used to evaluate the models on reconstruction error, factors of mean reconstruction and time to detection. A change point algorithm is required for the latter. Binary change point detection performs fast signal segmentation using a sequential approach: it identifies a change point in the signal, splits the series at that point, and repeats the process on the resulting sub-signals. This method offers low complexity, extends single change point detection to multiple points, and works without prior knowledge of the number of regimes.

For this use case, it detected similar change points to the more robust linearly penalized segmentation (PELT) [13] algorithm, and at a significantly faster rate; over 100 times faster in most cases. Only positive change points are considered for raising the alarm that a leak has occurred. Negative change points would indicate the fixing of a leak, but no such instances occur in our test dataset.

EVALUATION AND DISCUSSION

Reconstruction Error

As shown in Figure 2, the Windowed LSTM AE has the lowest reconstruction loss for the normal state. The deviation of the reconstruction errors from the median is also relatively low, indicating reliable reconstruction of normal data.

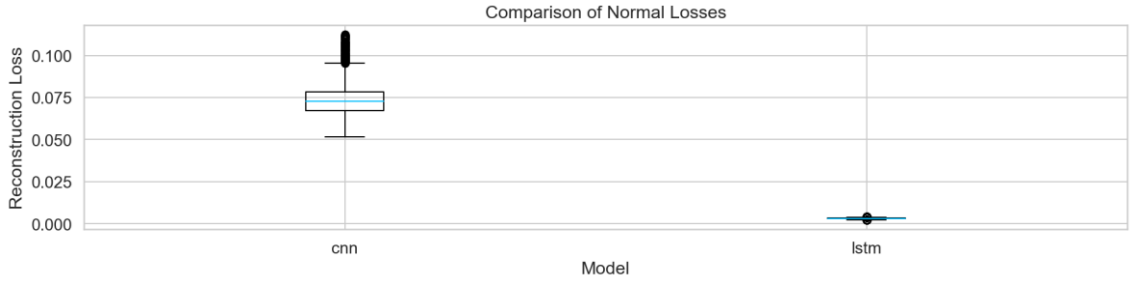


Figure 2. Reconstruction Error for Normal State Data

Factors of Mean Reconstruction Error

Table 1 represents the average reconstruction error for each architecture. As can be seen, the Windowed LSTM AE not only produced the lowest mean reconstruction error for the normal state, but also the highest deviation factors for both single leak and multi-leak situations.

Table 1. Factors of Mean Reconstruction Error

Model	Normal error	1 leak factor	2 leak factor
CNN AE	0.0732	2.4034	4.2370
LSTM AE	**0.0031**	**9.7775**	**16.9816**

Time to Detection (TTD)

Figure 3 shows that both abrupt and incipient leaks cause a significant rise in reconstruction error, aiding the Binary Segmentation change point detector. In DMA C, no leaks occurred from January 1 to February 9, 2019, resulting in minimal reconstruction error. Two leaks followed: an abrupt leak on February 10 at 13:05 and an incipient leak on May 30 at 21:55, both persisting until December 31. The change point detector successfully identified both leaks using reconstruction errors from both algorithms. For the CNN AE, a penalty of 10 and a period of a fortnight was selected, while for the LSTM AE, a penalty of 3 with a period of a month was used.

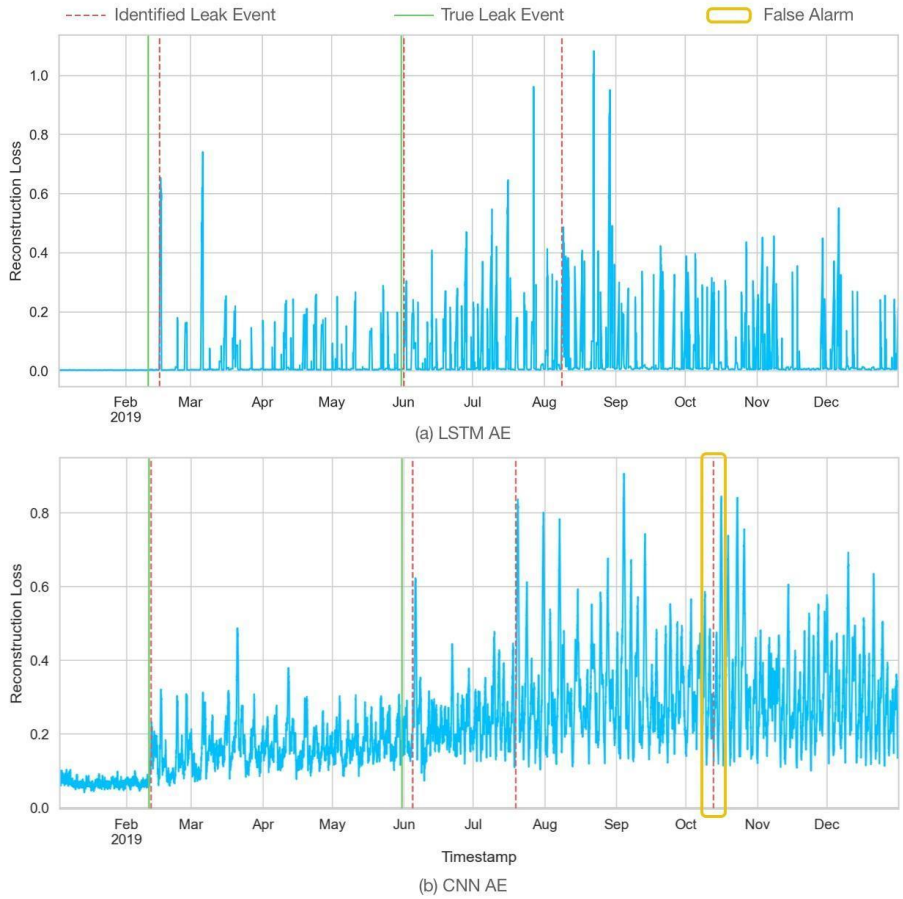


Figure 3. Comparison of TTD for Leak 1 (Abrupt) and Leak 2 (Incipient)

The CNN AE exhibited a detection lag of over four days for the incipient leak, whereas the LSTM AE detected it in under two days. Incipient leak detection depends on its growth rate—exponential growth leads to faster identification, while linear growth causes delays. The May 30 leak reached peak flow on August 11, therefore all interim alarms are considered part of the same event. A single false alarm was recorded for the CNN AE after this period.

Table 2. Time to Detection Comparison

Model	Abrupt leak (hh:mm)	Incipient leak (hh:mm)
1D CNN AE [3]	***12:40***	*93:50*
LILA-PINN [4]	*22:05*	*369:35*
CNN AE	21:25	117:20
LSTM AE	21:35	**43:55**

Table 2 shows the results for the two models and compares them with SOTA models from literature [3], [4]. The LSTM autoencoder significantly reduces the detection time for incipient leaks and performs comparably to other methods in detecting abrupt leaks, where a simple threshold-based approach is used.

CONCLUSION

The Windowed LSTM autoencoder demonstrated improved performance in detecting incipient leaks, achieving faster detection times while reducing false alarms. Future work could involve evaluating this method on additional datasets to enable broader comparisons against other established benchmarks.

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